

SAI PRANAV APPECHARLA VENKATA

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EDUCATION

University of Massachusetts Amherst

Amherst, MA

Bachelor of Science in Computer Science Minor in Business (Planned)

Sep. 2026 – May 2030

- Recipient of the **Chancellor's Award**, a \$72,000 merit scholarship awarded over four years for academic excellence.
- Relevant Coursework: Foundations of Programming, Calculus I, General Physics I, First-Year Seminar, Public Health.

EXPERIENCE

AI & Python Intern

Jun. 2025 – Jul. 2025

IIT Jammu Executive Programs

Hybrid

- Selected for a competitive pre-university research internship in Applied Artificial Intelligence under IIT Jammu faculty.
- Developed and evaluated machine learning models using real-world datasets, performing preprocessing, feature engineering, model training, and error analysis.
- Applied statistical analysis and machine learning techniques to solve practical research problems and presented findings during faculty review sessions.

Selected Delegate

Jun. 2023 – Jul. 2023

PASCH Exchange — Goethe-Institut

Bremen, Germany

- Awarded a fully funded place in the PASCH International Exchange Program representing India in Germany.
- Completed intensive German language instruction, earning the Goethe-Institut A2 certification while collaborating on multicultural team projects.

PROJECTS

CALIGO – Autonomous Personal AI Agent | Python, Claude Agent SDK, MCP

2026 – Present

- Developing an autonomous AI agent for planning, reasoning, and multi-step task execution.
- Integrating persistent memory, MCP servers, and modular tool calling.
- Building an evaluation framework for autonomy, consistency, and failure analysis.
- Studying limitations of LLM agents across context, memory, and tool use.

Detecting the Undetectable | Survey Paper

2026

- Surveying methods for detecting citation hallucinations in Large Language Models.
- Comparing benchmark evaluation with real-world document performance.
- Preparing an arXiv preprint in Natural Language Processing (cs.CL).

Claude Agent Experiments | Python, Claude Code, MCP

2026

- Developing experimental AI agents using Anthropic's Claude Agent SDK to explore autonomous planning, tool use, and memory management.
- Evaluating agent behavior under increasingly complex tasks to understand practical limitations of current LLM-based autonomous systems.

TECHNICAL SKILLS

Programming Languages: Python, Java

Artificial Intelligence: Claude Agent SDK, Claude Code, Model Context Protocol (MCP), LLM Agents, Prompt Engineering, Machine Learning, AI Evaluation

Cloud & Operating Systems: AWS Fundamentals, Linux, Cloud Computing

Cybersecurity: Ethical Hacking, Network Security, Kali Linux

Developer Tools: Git, GitHub, HTML, CSS, Linux CLI

Languages: English (Fluent), Telugu (Native), Hindi (Professional), German (A2), Japanese (Learning)

CERTIFICATIONS

Anthropic Academy

- Completed nine professional courses covering Claude Code, Model Context Protocol (MCP), Agent Skills, AI Fluency, and applied LLM development.

Google Cybersecurity Professional Certificate

Programming with Python & Exposure to AI — IIT Jammu Executive Programs

Cybersecurity & Ethical Hacking Workshop — IIIT Hyderabad

German A2 Certification — Goethe-Institut

Currently Pursuing

- AWS Certified Cloud Practitioner (CLF-C02)
- GitHub Foundations

HONORS

- Recipient of the **Chancellor's Award Scholarship** (\$72,000), University of Massachusetts Amherst.
- Selected for the IIT Jammu Executive Programs AI Research Internship.
- Selected Delegate for the PASCH International Exchange Program, Goethe-Institut Germany.